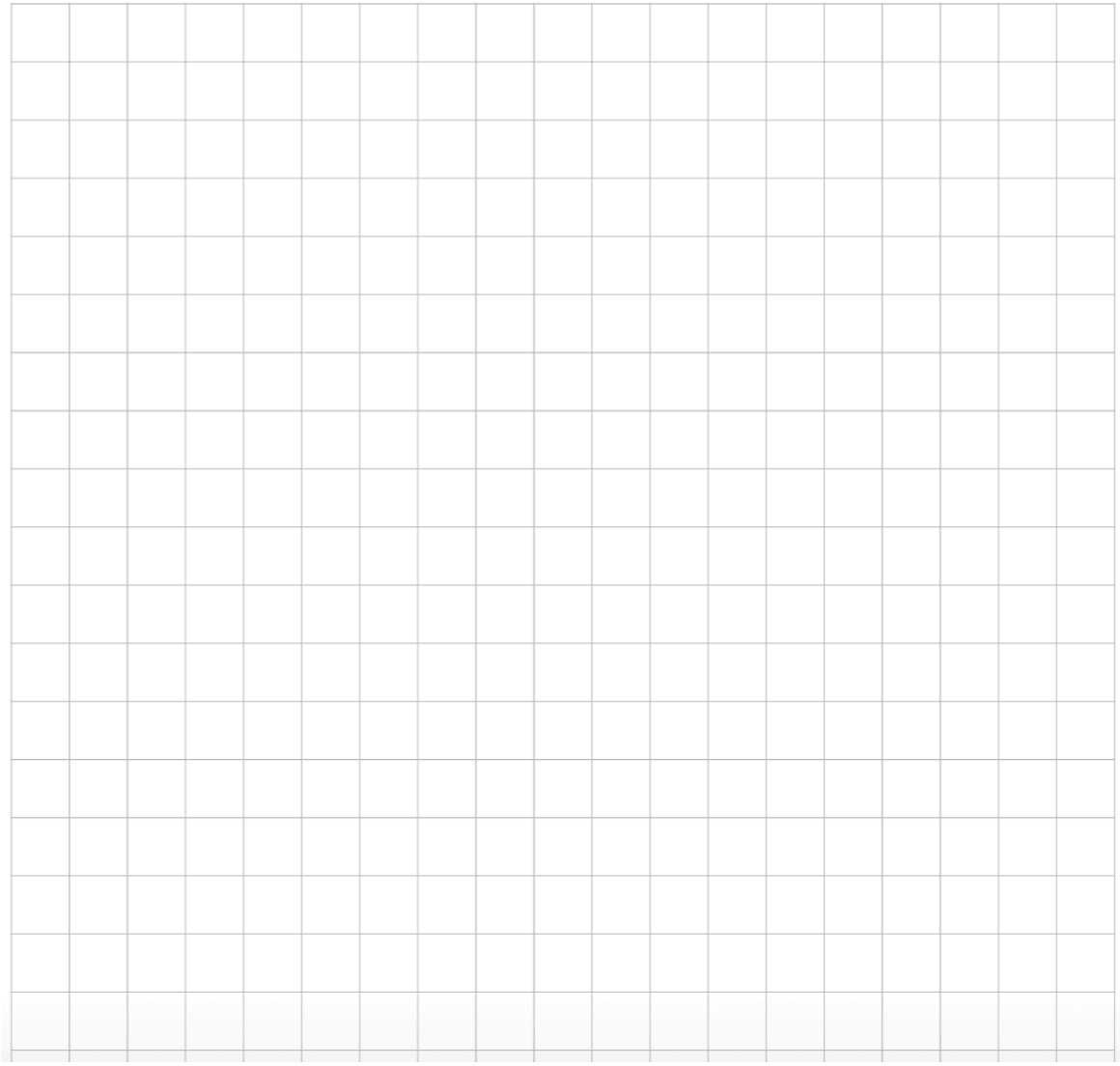
Matchstick Troubles

You have 48 matchsticks to lay on the grid below. the length of a matchstick is the length of one interval

Design a shape that has:
(a) The longest perimeter
(b) The greatest area